

StockBroker — Business Requirements Specification (as-is)

Generated 2026-10-04 from `Business-Requirements-Specification.md`

Contents

1. Document control
2. Executive summary
3. Product overview and scope
4. Actors and roles
5. Business glossary
6. Business information model
7. Business processes
 - P-001 — Place a trade
 - P-002 — Add funds
 - P-003 — Register a new client
8. Functional requirements by capability
 - 8.1 Account and access
 - 8.2 Wallet funding
 - 8.3 Market and pricing
 - 8.4 Trading
 - 8.5 Portfolio and dashboard
 - 8.6 Trade documents (invoices)
 - 8.7 Notifications (inbox)
9. Channel differences (portal vs public API)
10. Data protection and control behaviours
11. Known issues and risks (not requirements)
12. Open decisions for product
- Appendix A — Traceability
- Appendix B — Coverage
- Appendix C — How to give feedback

1. Document control

Item	Value
Document	Business Requirements Specification, as-is (reverse-engineered)
Version	0.1 (draft)
Date	2026-10-04
System	StockBroker demo platform: web portal and public REST API
Code version analysed	commit <code>ae607a6</code>
Generation mode	draft : includes requirements not yet confirmed
Generated by	/re-publish from projects/stockbroker/requirements/. Do not edit by hand. Correct the working files and regenerate.

Confirmation status

Confirmed	Changed	In review	Draft	Total	% confirmed
0	0	21	75	96	0%

⚠ **Read this first.** Every requirement in this document was derived from the system's source code. It describes **how the system behaves today**, not how anyone has agreed it *should* behave. Until product confirms an item, treat it as a statement of current behaviour. Items marked "In review" are in the current product review pack.

2. Executive summary

StockBroker is a **simulated retail stock-broking platform**. A client:

1. registers
2. funds a cash wallet by (simulated) bank transfer or debit card
3. buys and sells whole shares in 41 listed US companies at simulated, slightly fluctuating prices
4. receives a PDF invoice and inbox notifications for every trade

No real money, market data or payment network is involved, and the product says so on every money-related screen and document.

The platform is reached through two **channels**: a self-service **web portal**, and a **public REST API** for external client applications. The API offers most of the portal's operations, but not all of them.

Headline risks (section 11):

- A failure after a trade has already gone through can show the client an error, which invites a duplicate trade.
- Only the portal's button prevents duplicate orders. The platform itself doesn't detect them, and API callers have no protection at all.
- The trading rules are maintained twice, once per channel.

Key open decisions (section 12):

- price tolerance between the quote and the execution price
- whether the illustrative brokerage fee should be charged
- the KYC process
- who the public API is for

3. Product overview and scope

Purpose (inferred): a zero-risk sandbox that behaves like a real broker's core loop: fund → browse → trade → receipt → history.

Channels

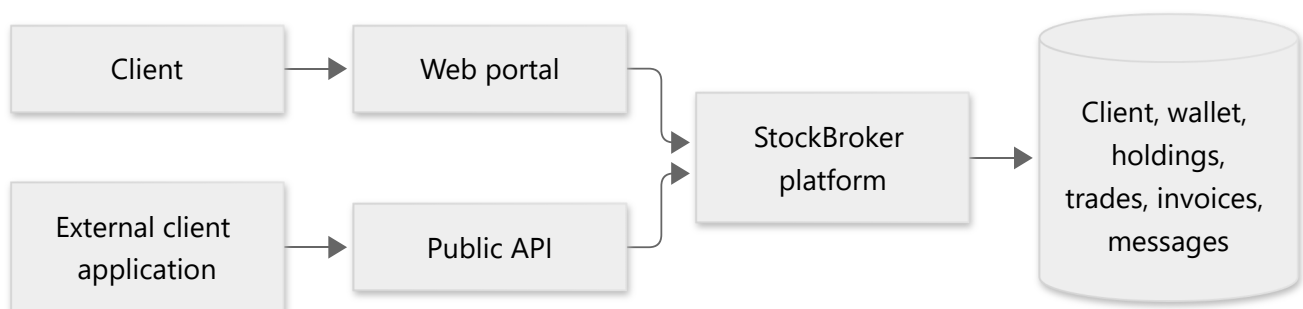
Channel	Users	Notes
Web portal	Clients (self-service)	Full feature set
Public REST API	External client applications (consumers unknown, see Q-API-001)	Token-based. No transaction history, read-status or logout.

In scope today

- account registration, login and profile
- wallet funding
- company catalogue and prices
- buy and sell orders with immediate execution
- portfolio valuation and dashboard
- trade invoices
- inbox notifications

Not present in the system

- back-office or operations functions (no admin role)
- withdrawals
- order types other than immediate buy/sell (limit, stop, cancel)
- fees charged on trades
- realised profit and loss
- KYC verification
- password reset
- batch or end-of-day processing
- connections to real markets, banks or card networks



4. Actors and roles

Actor	Description	Can do
Visitor	Not signed in	Register, log in
Client	Signed-in account holder	Everything in sections 8.1–8.7, for their own data only (BR-WEB-004)
External client application	Software acting for a client, with the client's access token	The API subset (section 9)
Operations / administrator	—	No such role exists (Q-FO-003)

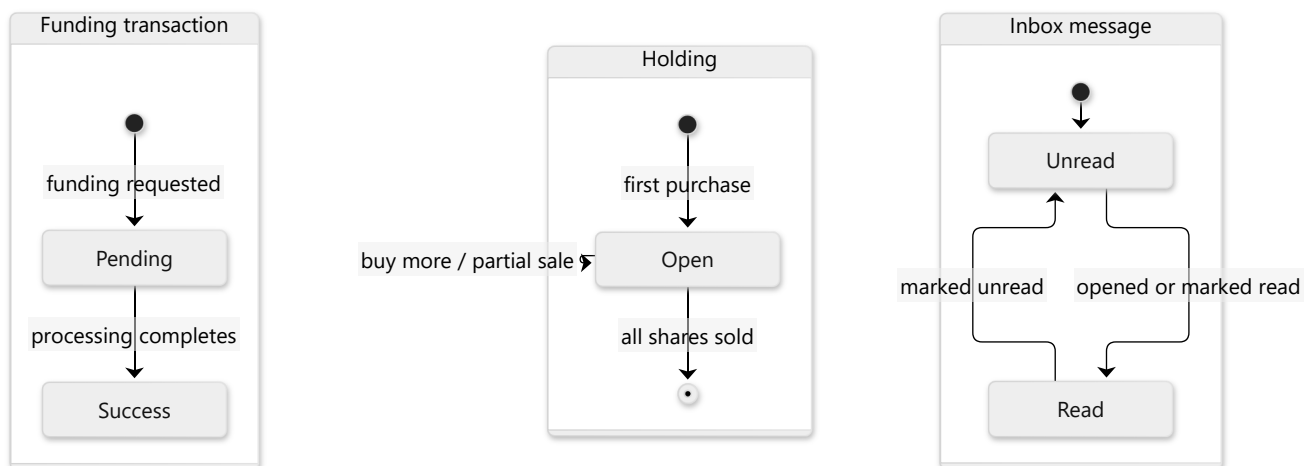
5. Business glossary

Term	Meaning
Client	A person registered on the platform
KYC status	Know-your-customer state: Unverified, Pending or Verified. Displayed only. Never changes and is never checked.
Wallet	The client's single USD cash account. It is the only source of buying power.
Wallet transaction	A credit or debit to the wallet, from funding or from a trade
Funding	Adding cash to the wallet by bank transfer or debit card (simulated)
IFSC-style branch code	Bank branch identifier: 4 letters, a zero, then 6 letters or digits
Company / ticker / sector	A tradable listed company, its exchange symbol, and its industry group
Base price	Reference share price, around which the simulated price moves
Market price (simulated)	Base price $\pm 2\%$, newly drawn every time a price is shown or used
Quoted price	The price shown while building an order. It is not guaranteed.
Execution price	The price at which the trade actually executes, set at confirmation
Trade	An executed buy or sell of whole shares of one company
Trade total	Execution price $\times$ quantity, rounded to the cent
Holding	Shares of one company that a client owns, with their average cost
Average cost	Weighted average purchase price per share
Portfolio value	Current market value of all holdings
Unrealised gain/loss	Current value minus cost of current holdings
Invoice	The document issued for every trade (PDF)
Inbox message	An in-app notification: System or Invoice
Session	The signed-in period: 7 days
Demo mode	The platform-wide notice that no real money or trades are involved

6. Business information model

Entity	Definition	Key facts
Client	A registered person	Email is unique and case-insensitive. Email and KYC can't be changed by the client.
Wallet	The client's cash account	Exactly one per client. Opens at \$0. Never goes negative through trading.
Wallet transaction	One cash movement	Type: credit or debit. Method: bank transfer, debit card or trade. Status: Pending, Success or Failed (Failed is never used).
Company	Tradable company (41, reference data)	No add, change or delist function
Holding	Shares owned in one company	One per client per company. Removed when it reaches zero.
Trade	An executed order	Always recorded as Completed. Failed trades leave no record.
Invoice	Trade document	One per trade. Number format "INV-" + 8 characters.
Inbox message	Notification	Starts unread. Can be toggled read/unread. Can't be deleted.

Lifecycles



7. Business processes

P-001 — Place a trade

Goal: the client buys or sells whole shares. Cash and shares move immediately, and the trade is documented.

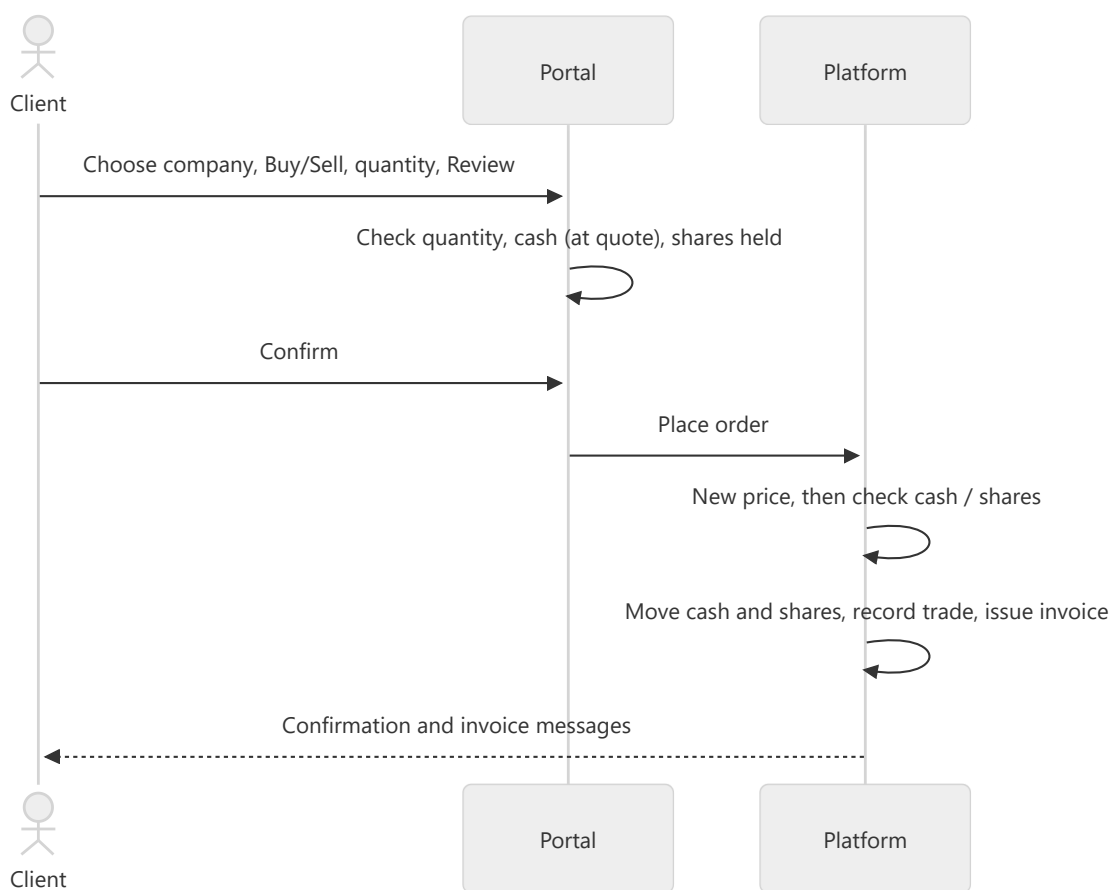
Trigger: the client confirms an order in the portal, or an external application submits one through the API.

1. The client picks a company and chooses Buy or Sell. The screen shows the quoted price.
2. The client enters a quantity and reviews the order. The portal checks the quantity, the cash (at the quoted price) and the shares held.
3. The client confirms. The confirm button is disabled until the result arrives.
4. The platform takes a **new** price at that moment. This is the execution price.
5. **Buy:** the platform checks the cash, takes the total from the wallet and adds the shares to the holding at a weighted average cost. **Sell:** the platform checks the shares held, reduces or removes the holding and adds the total to the wallet.

- The trade is recorded as Completed with an invoice. All money and share changes happen together, or not at all.
- The client receives a "Trade executed" message and an invoice message.

Outcomes when the order doesn't go through

Situation	What the client sees	Effect
Invalid quantity, or not enough cash or shares at review	Message on screen	Nothing sent
Not enough cash at the execution price	"Insufficient wallet balance for this order"	Nothing changes
Not enough shares (e.g. sold elsewhere meanwhile)	"Insufficient holdings for this order"	Nothing changes
Platform busy or rare internal error	"Internal server error"	Nothing changes
Notification fails after the trade completed	"Internal server error"	The trade went through. A retry would trade again (section 11).



P-002 — Add funds

Goal: the client adds cash to the wallet.

- The client chooses bank transfer or debit card and enters the amount and details. The portal checks them.
- The platform checks them again and records the deposit as **Pending**. It keeps only the method and the last 4 digits.

3. After a short simulated processing time (bank transfer about 1.5 s, card about 0.8 s), the wallet is credited and the deposit becomes **Success**.
4. The client receives "Funds added successfully" with the new balance.

Exceptions:

- Funding never fails once the details are valid. There is no declined path.
- If the platform stops during processing, the deposit stays Pending forever (section 11).

P-003 — Register a new client

1. The visitor enters name, email, phone and password, and the details are checked.
2. Registration is refused if the email is already registered.
3. The account (KYC Unverified) and a \$0 wallet are created together.
4. A welcome message is sent.
5. The client is signed in immediately, with no email verification, and taken to the dashboard.

8. Functional requirements by capability

Channel: Portal / API / Both · **Status:** Draft / In review / Confirmed / Changed

8.1 Account and access

ID	Requirement	Channel	Status
BR-WEB-001	Every operation except registering, logging in and logging out requires the client to be signed in.	Both	Draft
BR-FO-001	Visitors who aren't signed in can reach only the login and registration pages. Signed-in clients are sent to the dashboard.	Portal	Draft
BR-WEB-002	After logging in or registering, a client stays signed in for 7 days. There is no inactivity timeout.	Both	Draft
BR-API-001	API clients receive an access token (valid 7 days) at registration or login and must present it on every call.	API	Draft
BR-API-002	Any client application may call the public API. △ unverified: the intended consumers are unknown.	API	Draft
BR-WEB-003	A request with an invalid or expired session is refused.	Both	Draft
BR-WEB-007 (also BR-DB-001)	Registration is refused if an account already exists with the same email. Emails are case-insensitive.	Both	Draft
BR-LIB-008	A password must be at least 8 characters and contain at least one letter and one digit.	Both	Draft
BR-LIB-009	Name is required, at most 100 characters.	Both	Draft
BR-LIB-010	Email must be valid. It is stored in lower case.	Both	Draft
BR-LIB-011	Phone: 7–20 characters, using digits, spaces, "+", "-", "(" and ")".	Both	Draft
BR-LIB-012	Address is optional (at most 200 characters) and can only be set in the profile, not at registration.	Both	Draft
BR-WEB-008 (also BR-DB-007)	Every new client gets a wallet with zero balance at registration.	Both	Draft

ID	Requirement	Channel	Status
BR-DB-002	Each client has at most one wallet.	Both	Draft
BR-DB-006	Every new client starts with KYC status Unverified.	Both	Draft
BR-WEB-009	A new client receives a welcome inbox message.	Both	Draft
BR-WEB-010	A successful registration signs the client in immediately. There is no email verification.	Both	Draft
BR-WEB-011	A failed login shows the same message for an unknown email or a wrong password. There is no limit on attempts.	Both	Draft
BR-WEB-012	Logging out ends the session in that browser.	Portal	Draft
BR-WEB-013	A client can change name, phone and address only. Email, KYC status and member-since date are fixed.	Both	Draft
BR-WEB-014	A client can clear their address.	Both	Draft
BR-FO-002	The registration form checks the fields before submitting and shows server messages against the field.	Portal	Draft
BR-FO-003	After login or registration, the client lands on the dashboard.	Portal	Draft
BR-FO-013	The profile shows email, member-since date and KYC status (labelled mock) as read-only. Name, phone and address are editable.	Portal	Draft

8.2 Wallet funding

ID	Requirement	Channel	Status
BR-LIB-013	A funding amount must be above 0 and at most 1,000,000 per transaction. There is no daily or cumulative limit.	Both	Draft
BR-LIB-014	Bank transfer: the account number must be 6–18 digits.	Both	Draft
BR-LIB-015	Bank transfer: the branch code must be in IFSC style (e.g. DEMO0123456). Lower case is accepted.	Both	Draft
BR-LIB-016	Debit card: the number must be 13–19 digits and pass the standard card checksum.	Both	Draft
BR-LIB-017	Debit card: the expiry must be in MM/YY format and not earlier than the current month.	Both	Draft
BR-LIB-018	Debit card: the CVV must be 3 or 4 digits.	Both	Draft
BR-WEB-020	A deposit is recorded as Pending, then credited in full and marked Success after the simulated processing time. Funding never fails once the details are valid.	Both	Draft
BR-WEB-021 (also BR-LIB-020)	Only the method and the last 4 digits of the account or card are kept. Full numbers, branch code, expiry and CVV are not stored.	Both	Draft
BR-WEB-022	After a successful deposit, the client receives a message with the amount, the method and the new balance.	Both	Draft
BR-WEB-019	The wallet history lists all deposits and trade movements, newest first.	Portal	Draft
BR-FO-010	The funding form checks the details before submitting.	Portal	Draft
BR-FO-011	The Add-funds button is disabled while processing. On success it shows the amount and the new balance.	Portal	Draft
BR-FO-012	The history shows date, description, method, status and a signed amount.	Portal	Draft
BR-API-007	Funding through the API returns only the new balance.	API	Draft

8.3 Market and pricing

ID	Requirement	Channel	Status
BR-LIB-001	A share's price is its base price moved randomly by up to $\pm 2\%$, rounded to the cent, and drawn anew every time it is needed.	Both	In review
BR-DB-008	Each company has a unique ticker.	Both	Draft
BR-WEB-015	Companies can be searched by any part of the ticker, name or sector. They are listed alphabetically by ticker.	Both	Draft
BR-WEB-016	The company list shows a current simulated price for each company.	Both	Draft
BR-FO-008	The company list refreshes its prices every 15 seconds.	Portal	Draft

8.4 Trading

ID	Requirement	Channel	Status
BR-LIB-019	An order must name a company, be Buy or Sell, and be for at least 1 whole share. There is no maximum size.	Both	In review
BR-WEB-024	An order for an unknown company is refused.	Both	In review
BR-WEB-025	The trade executes at a price taken at confirmation, not the price the client saw. There is no limit on the difference (up to about 4%).	Both	In review
BR-WEB-026	Trade total = execution price $\times$ quantity, rounded to the cent.	Both	In review
BR-WEB-027	A buy is refused if the wallet balance is less than the total. Exactly enough is allowed.	Both	In review
BR-WEB-028	A buy debits the wallet and adds the shares to the holding at a weighted average cost. A new holding's average cost is the execution price.	Both	In review
BR-WEB-029	A sell is refused if the client holds fewer shares than requested. Short selling isn't possible.	Both	In review
BR-WEB-030	A sell reduces the holding and removes it at zero. The average cost of the remaining shares is unchanged.	Both	In review
BR-WEB-031	A sell credits the full total to the wallet immediately. There is no settlement delay.	Both	In review
BR-WEB-032	Every trade adds a wallet movement: a debit for a buy, a credit for a sell.	Both	In review
BR-WEB-033	Every trade is recorded as Completed and receives an invoice at the same moment.	Both	In review
BR-WEB-034	All money and share changes of a trade happen together, or not at all.	Both	In review
BR-WEB-035	After every trade the client receives a confirmation message and an invoice message.	Both	In review
BR-WEB-036	No brokerage, commission or tax is charged on trades.	Both	In review

ID	Requirement	Channel	Status
BR-DB-003	A client's shares in a company are always one holding.	Both	Draft
BR-FO-004	Every company offers Buy and Sell, even when the client holds none of it.	Portal	In review
BR-FO-005	Review refuses an invalid quantity, a buy above the wallet balance (at the quote), and a sell above the shares held.	Portal	In review
BR-FO-006	Orders take two steps, Review then Confirm. Confirm warns that the final price may differ slightly.	Portal	In review
BR-FO-007	The confirm button is disabled while the trade executes, so a double-click can't trade twice.	Portal	In review
BR-FO-009	The success message shows the executed price and total.	Portal	Draft
BR-WEB-023	The trade history lists all trades, newest first, with invoice links.	Both	Draft

8.5 Portfolio and dashboard

ID	Requirement	Channel	Status
BR-LIB-002	Only holdings with at least one share are shown.	Both	Draft
BR-LIB-003	Each holding shows: current value = price × quantity; gain/loss = value – average cost × quantity; gain/loss % (0% if the cost is zero).	Both	Draft
BR-WEB-017	The dashboard shows wallet balance, portfolio value, total gain/loss with %, holdings, and the 5 most recent trades.	Portal	Draft
BR-WEB-018	Total gain/loss % = total gain/loss ÷ total cost of holdings. It is 0% when the client holds nothing.	Portal	Draft
BR-API-003	The API portfolio summary shows cash, stock value, total account value (cash + stock) and holdings.	API	Draft

8.6 Trade documents (invoices)

ID	Requirement	Channel	Status
BR-LIB-004 (also BR-DB-005)	Invoice numbers are "INV-" + 8 characters taken from the trade reference, and are unique.	Both	In review
BR-DB-004	A trade has at most one invoice.	Both	Draft
BR-LIB-005	The invoice shows number, date, the client's name, email, phone and address, the trade line (type, company, quantity, price, amount) and the total.	Both	Draft
BR-LIB-006	The invoice shows a brokerage fee of 0.1% (min \$0.50), labelled "informational only — not charged".	Both	In review
BR-LIB-007	Every invoice states that it is a simulated demo document with no legal or financial validity.	Both	Draft
BR-WEB-038	Only the client who placed the trade can open its invoice.	Both	Draft
BR-API-004	API clients can look up an invoice by trade reference, with a download link.	API	Draft
BR-WEB-037	A client can list all their invoices. △ unverified: no screen uses this.	Portal	Draft

ID	Requirement	Channel	Status
BR-FO-016	Invoices open from trade history, from recent trades on the dashboard, and from the invoice message.	Portal	Draft

8.7 Notifications (inbox)

ID	Requirement	Channel	Status
BR-DB-009	New messages start unread.	Both	Draft
BR-WEB-039	The inbox lists all messages, newest first. Messages can't be deleted.	Both	Draft
BR-WEB-040	The unread count = messages not yet read.	Portal	Draft
BR-WEB-041	A client can mark a message read or unread.	Portal	Draft
BR-FO-014	Opening the inbox shows the newest message. Viewing a message marks it read.	Portal	Draft
BR-FO-015	The unread badge refreshes every 30 seconds and is hidden at zero.	Portal	Draft
BR-API-005	Reading a message through the API doesn't mark it read.	API	Draft
BR-API-006	For trades placed via the API, the invoice message points to the API lookup instead of trade history.	API	Draft
BR-FO-017	A "Demo Mode — no real money" notice appears on every signed-in page and on the login, registration and payments screens.	Portal	Draft

9. Channel differences (portal vs public API)

Capability	Portal	Public API
Sign-in	Session (7 days)	Access token (7 days)
Logout	Yes	No
Dashboard	Yes	No. A portfolio summary with total account value instead.
Wallet history	Yes	No
Invoice access	From trades and inbox	Lookup by trade reference
Mark message read / unread count	Yes	No. Reading leaves messages unread.
Duplicate-order protection	Button disabled while executing	None
Invoice message wording	Points to trade history	Points to the API lookup
Callers	The portal only	Any application

All trading, funding and registration rules apply identically in both channels. They are implemented twice (section 11, D-API-001).

10. Data protection and control behaviours

ID	Behaviour
BR-WEB-042	Passwords are stored only as a one-way hash and never returned.
BR-WEB-021	Card and bank details are never stored; only the last 4 digits are kept.
BR-WEB-004	Clients can see and change only their own data. There are no other roles.
BR-WEB-005	Another client's invoice or message is reported as "not found".
BR-WEB-006	Invalid input is rejected with a message per field.
BR-WEB-034	Trade money and share movements are all-or-nothing.
BR-DB-010	Records with history can't be deleted.

11. Known issues and risks (not requirements)

Observations from the code analysis, written as business impact. They need product and engineering decisions.

High impact

ID	Issue	Business impact	Decision
D-WEB-002	Trade and funding confirmations are sent after the operation has been saved, separately	If sending fails, the client sees an error for a trade or deposit that succeeded, and may repeat it	Open
D-WEB-004 / D-API-005	No platform-level protection against duplicate orders or deposits	Retries, or API callers, can create duplicate trades or deposits	Open
D-API-001	Trading, funding and registration rules are maintained twice (portal and API)	Changes must be made twice. The channels can drift apart (they already differ in message wording).	Open

Medium impact

ID	Issue	Business impact	Decision
D-WEB-003	A deposit interrupted during processing stays Pending forever. Nothing reconciles it.	Client's money is "lost" in Pending, with no recovery process	Open
D-WEB-001 / D-API-002	Logout doesn't invalidate the session server-side; the API has no logout	A leaked session or token works for up to 7 days	Open
D-WEB-007	No limit on failed login attempts	Password guessing is possible	Open
D-DB-001	Money is stored as floating-point numbers	Cent-level rounding drift over time	Open
D-DB-002	Portal and API write to the same small database at the same time	Under load, trades or deposits can fail with a generic error	Open
D-LIB-001	Invoice numbers use only 8 characters of the trade reference	A rare collision makes the trade fail	Open
D-DOM-001	Statuses exist that are never used (Failed trade, Failed deposit, KYC Pending/Verified)	Failed operations leave no audit trail. Screens imply states that can't happen.	Open

Low impact

ID	Issue	Business impact	Decision
D-WEB-006 / D-FO-003	The portal checks cash at the quote, which may be stale; the platform checks at the execution price	Occasional "insufficient balance" after a passed review	Open
D-FO-001 / D-FO-002	The portal checks phone and funding maximum less strictly than the platform	Errors appear only after submitting	Open
D-API-003	Messages read via the API stay unread in the portal	Inconsistent unread badge	Open
D-API-004	The published API contract states a funding minimum of 0.01; the platform accepts smaller amounts	Misleads integrators	Open
D-WEB-005	A "list my invoices" function exists but no screen uses it	Dead function, or a missing screen	Open

12. Open decisions for product

Capability	ID	Decision needed
Account	Q-DOM-001 / Q-WEB-005	How does KYC progress, and should funding or trading require it?
Account	Q-DOM-005	Can clients close their account?
Account	Q-WEB-004	Is a fixed 7-day session acceptable, or is an inactivity timeout needed?
Account	Q-FO-002	Are forgot-password and change-password required?
Account	Q-FO-003 / Q-WEB-007	Is an operations role needed? Is an audit trail required?
Funding	Q-DOM-002	Should withdrawals exist?
Funding	Q-DOM-003	When should a deposit fail, and what should the client see?
Funding	Q-LIB-001	Daily or monthly limits? A minimum deposit?
Market	Q-DOM-004	Who maintains the company list? Can a company be suspended?
Market	Q-WEB-003	Should search ignore upper/lower case?
Trading	Q-WEB-001	Maximum allowed difference between quote and execution price?
Trading	Q-LIB-002	Should the 0.1% brokerage fee be charged?
Trading	Q-WEB-002	Should refused or failed trades be recorded?
Trading	Q-WEB-006	Should realised profit/loss be calculated?
Trading	Q-PRC-001	Immediate settlement and 24/7 trading, or settlement cycles and market hours?
Trading	Q-FO-001	Hide Sell for companies the client doesn't hold?
Documents	Q-LIB-003	Should invoices keep the client details as at trade time?
API	Q-API-001	Who is the public API for? Should it match the portal's features?
API	Q-API-002	Should a portal session work on the API and vice versa?

Appendix A — Traceability

Every requirement ID resolves to a working file in `projects/stockbroker/requirements/`, which cites the exact code location (231 citations, all checked).

ID prefix	Working file	Requirements
BR-DB	03-capabilities/database/rules.md	10
BR-LIB	03-capabilities/core-library/rules.md	20
BR-WEB	03-capabilities/web-server/rules.md	42
BR-API	03-capabilities/public-api/rules.md (also maps 29 duplicated WEB rules)	7
BR-FO	03-capabilities/front-end/rules.md	17
P-001...003	02-processes/	3 processes

Appendix B — Coverage

Component	Analysed	Independently verified
Web portal (screens)	Yes	Pending
Portal back end	Yes	Same-session check only (1 correction)
Public API	Yes	Pending
Shared domain library	Yes	Pending
Database	Yes	Pending

Appendix C — How to give feedback

Product decisions are captured through **review packs** (`requirements/_review/`), one per process. Mark each rule:

Decision	Meaning
Confirmed	Correct as described
Changed	Write the correct wording
Rejected (bug)	The system shouldn't do this
Obsolete	No longer needed
Unsure	Needs discussion

Decisions are applied to the working files with `/re-apply-review`, and this document is then regenerated with `/re-publish`. Once enough is confirmed, publish with `/re-publish baseline` to produce the agreed requirements baseline.